Research Plan: COD Prediction in Wastewater Treatment

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Core Question: How do different temporal feature engineering strategies (Raw Sensor Baseline vs. Kalman Filtering vs. Kalman+ARIMA) and different modelling architectures (Traditional Machine Learning vs. Deep Learning) affect out-of-sample R^2 performance when predicting wastewater `AT_COD`?

I. Introduction

II. Materials and Methods

A. Dataset and Biological Treatment Process

To maintain industrial confidentiality, the dataset used in this study was derived by applying a controlled stochastic perturbation to operational measurements from an active biological wastewater treatment facility. Specifically, independent Gaussian noise with a standard deviation of 0.5% of each variable's observed range was applied to the original sensor readings, producing a statistically representative simulated dataset that preserves the temporal dynamics and inter-variable relationships of the real process while preventing proprietary data disclosure. The resulting dataset comprised 983 chronological daily observations spanning the full operational record of the facility.

The treatment process follows a four-stage pipeline, illustrated in Fig. 1. Raw industrial wastewater first enters the Process Wastewater Tank (PWT), which acts as a primary collection and homogenisation buffer. The wastewater then passes through Wet Air Oxidation (WAO), a thermochemical pre-treatment stage that breaks down refractory organic compounds prior to biological treatment. The partially oxidised effluent subsequently enters the Equalization Tank (ET), which stabilises flow and load variability before the final biological stage. The aeration tank (AT) constitutes the core activated sludge reactor, where aerobic microorganisms mineralise residual organic matter under controlled aeration.

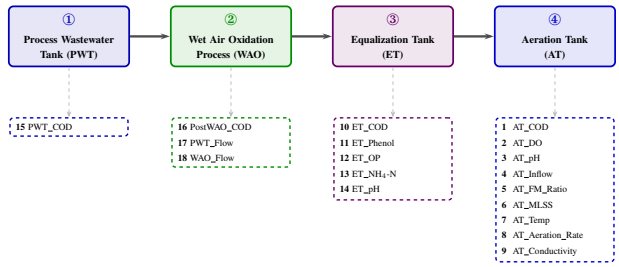

Fig. 1. Four-stage industrial wastewater treatment pipeline. Solid arrows denote the direction of wastewater flow. Dashed arrows indicate the sensor variables collected at each stage; `AT_COD` (1) and `AT_DO` (2) are the primary prediction targets, and all remaining variables serve as input features.

The two primary prediction targets in this study are the Chemical Oxygen Demand within the aeration tank (AT_COD), which quantifies the residual organic load at the biological treatment stage, and the Dissolved Oxygen concentration (AT_DO), which governs aerobic microbial activity and process efficiency. The remaining variables across all four process stages serve as input features. Table I summarises the full set of measured variables, their abbreviations, and their roles in the predictive modelling pipeline.

TABLE I
Measured Process Variables Grouped by Treatment Stage

	Variable Item	Abbreviation	Description
Aeration Tank Measurements	Aeration Tank COD	AT_COD	Predicted Target
	Aeration Tank Dissolved Oxygen	AT_DO	Predicted Target
	Aeration Tank Inflow Volume	AT_Inflow	Feature
	Aeration Tank Food-to-Microorganism Ratio	AT_FM_Ratio	Feature
	Aeration Tank Mixed Liquor Suspended Solids	AT_MLSS	Feature
	Aeration Tank Temperature	AT_Temp	Feature
	Aeration Tank Aeration Rate	AT_Aeration_Rate	Feature
	Aeration Tank Conductivity	AT_Conductivity	Feature
Equalization Tank Measurements	Equalization Tank COD	ET_COD	Feature
	Equalization Tank Phenol	ET_Phenol	Feature
	Equalization Tank Orthophosphate	ET_OP	Feature
	Equalization Tank Ammonia Nitrogen	ET_NH_4-N	Feature
	Equalization Tank pH	ET_pH	Feature
Process Measurements	Process Wastewater Tank COD Concentration	PWT_COD	Feature
	Post-Wet Oxidation COD Concentration	PostWAO_COD	Feature
	Process Wastewater Tank Direct Flow	PWT_Flow	Feature
	Wet Air Oxidation Process Flow	WAO_Flow	Feature

A fundamental physical constraint governing this process is the aeration tank's hydraulic retention time of approximately six days. Consequently, influent quality

fluctuations required up to six days to manifest in the effluent, meaning contemporaneous sensor readings did not reflect instantaneous cause-and-effect relationships. This temporal delay dictated the time-series architecture of the predictive models: input features were constructed as six-day rolling averages of the process variables to forecast the aeration tank state six days ahead, strictly aligning the machine learning formulation with the physical dynamics of the system.

B. Target-Derived Feature Engineering (The Kalman/ARIMA Pipeline)

C. Proposed Regression Architecture: Extra Trees

D. Deep Learning Benchmark (Phase 3 LSTM Optimization)

E. Experimental Setup and Evaluation Metrics

III. Results and Discussion

IV. Conclusion

V. References

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. 2019. Optuna: A Next-generation Hyperparameter Optimization Framework. In KDD.